NAU · DEPARTMENT of MATHEMATICS and STATISTICS

Problem of the Week

#4

Fall 2026

Due Monday, October 19 at 11:59 pm (Arizona time)

Two, Three, Six

Let a, b, and c be real numbers with $a \neq 0$, and suppose the equation

$$ax^2 + bx + c = 0$$

has no real solutions. Prove that

$$a\,(2a + 3b + 6c) > 0.$$

How to enter

Scan the code, or go to **ac.nau.edu/lms-apps/self-enroll/545026**, to join the POTW Canvas course. Then upload **one PDF** (typed, or neatly handwritten and scanned) to this week's assignment. Open to all NAU undergraduates.

Join on Canvas

The rules

- **Your own work only.** No AI tools (ChatGPT, Gemini, Claude, Copilot, Photomath, …), no internet, no calculators or software, and no help from other people.
- **Explain everything.** A bare answer earns little credit; a clear, complete argument earns full credit.
- Any participant may be asked to explain their solution in person.

Scored 0–4. Full rules, past problems, and model solutions:
naumathstat.github.io/problem-of-the-week
Questions: **ian.williams@nau.edu**